

Summary Table

Sample	Noise Types (search)	Dose Scale	Architecture	Distinct Feature
sample_0083	speckle, raster_drift, shot_noise, edge_brightening	1.34	DRAM + FinFET	3-zone dot/band/fin composite
sample_0238	raster_drift, shot_noise, edge_brightening	1.23	Logic/BEOL	Alignment cross fiducial
sample_0429	salt_pepper, raster_drift, shot_noise, edge_brightening	1.19	DRAM (staggered)	Asymmetric defect blob
sample_0856	charging, speckle, raster_drift, shot_noise, edge_brightening	0.78	Logic/BEOL	Charging scan-line streaks
sample_1057	charging, speckle, salt_pepper, raster_drift, shot_noise, edge_brightening	0.55	DRAM + FinFET	Full 6-noise stack, wordline straps
sample_1393	charging, speckle, raster_drift, shot_noise, edge_brightening	0.38	Logic/BEOL (ring/via)	Salt-pepper present even in reference

Sample 1 — sample_0083: DRAM + FinFET (Legacy Style)

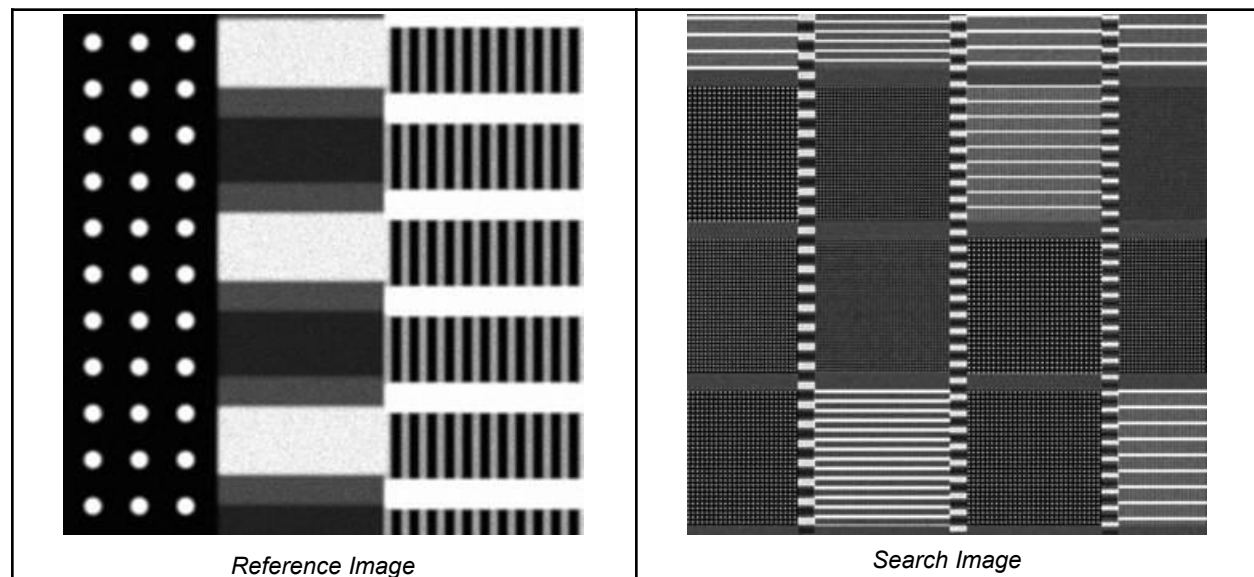
Difficulty Tier	Easy
Noise Variation (types)	speckle + raster_drift + shot_noise + edge_brightening
Dose Scale	1.344
Shear	0.599 px
Jitter	0.116 px
Beam Spot	4.483 nm
Edge Strength	0.424
Architecture Type	DRAM (dot-array capacitor region) + FinFET (vertical parallel-fin region) — legacy_dram_1x + legacy_finfet styles

Noise Citations	speckle: Boyat, A.K. & Joshi, B.K. (2015). "A Review Paper: Noise Models in Digital Image Processing." arXiv:1505.03489. https://arxiv.org/pdf/1505.03489 raster drift: Sang, X. & LeBeau, J.M. (2016). "Correcting nonlinear drift distortion of scanning probe microscopy from image pairs with orthogonal scan directions." Ultramicroscopy 187, 98-108 (arXiv:1507.00320). https://arxiv.org/pdf/1507.00320 shot noise: Sim, K.S. et al. (2025). "Signal-to-Noise Ratio in Scanning Electron Microscopy: A Comprehensive Review." arXiv:2510.07886. https://arxiv.org/abs/2510.07886 edge brightening: Electron Microscopy ETH Zurich. "Secondary Electron Imaging — Edge Effect." ETH Zurich Scientific Center for Optical and Electron Microscopy. https://www.microscopy.ethz.ch/se.htm
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Pattern Uniqueness — Justification

Three distinct sub-zones sit side by side: a regular dot grid representing DRAM cell contacts, a horizontally banded gray-level ramp, and dense vertical fin stripes representing FinFET gates. No other sample in this set combines all three zone types with this low-jitter, speckle-dominant noise signature — the low shear (0.6px) and jitter (0.12px) keep the geometry crisp while grain noise is still clearly visible.

Reference & Search Images



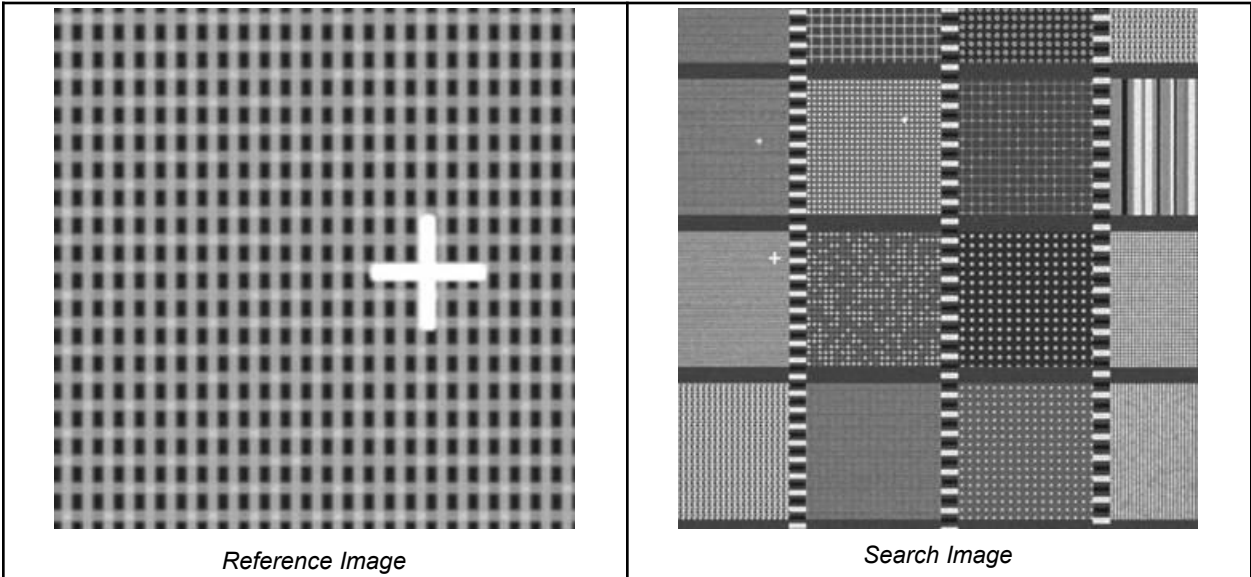
Sample 2 — sample_0238: Logic / BEOL Mesh with Alignment Marker

Difficulty Tier	Easy
Noise Variation (types)	raster_drift + shot_noise + edge_brightening (no speckle / salt-pepper)
Dose Scale	1.229
Shear	0.828 px
Jitter	0.374 px
Beam Spot	5.144 nm
Edge Strength	0.421
Architecture Type	Logic / BEOL interconnect (mesh_grid, contact_array_square, std_cell) — not DRAM or FinFET; a metal-line crosshatch mesh
Noise Citations	raster drift: <i>Sang, X. & LeBeau, J.M. (2016). "Correcting nonlinear drift distortion of scanning probe microscopy from image pairs with orthogonal scan directions." Ultramicroscopy 187, 98-108 (arXiv:1507.00320).</i> https://arxiv.org/pdf/1507.00320 shot noise: <i>Sim, K.S. et al. (2025). "Signal-to-Noise Ratio in Scanning Electron Microscopy: A Comprehensive Review." arXiv:2510.07886.</i> https://arxiv.org/abs/2510.07886 edge brightening: <i>Electron Microscopy ETH Zurich. "Secondary Electron Imaging — Edge Effect." ETH Zurich Scientific Center for Optical and Electron Microscopy.</i> https://www.microscopy.ethz.ch/se.htm

Pattern Uniqueness — Justification

This is the only sample with a plus-shaped white alignment/fiducial cross breaking an otherwise perfectly periodic crosshatch mesh — a landmark feature none of the other five samples have. It also carries the minimal 3-type noise stack (no speckle or salt-pepper), making it the cleanest sample in the set.

Reference & Search Images



Sample 3 — sample_0429: DRAM
Staggered Bitline (Realistic Style)

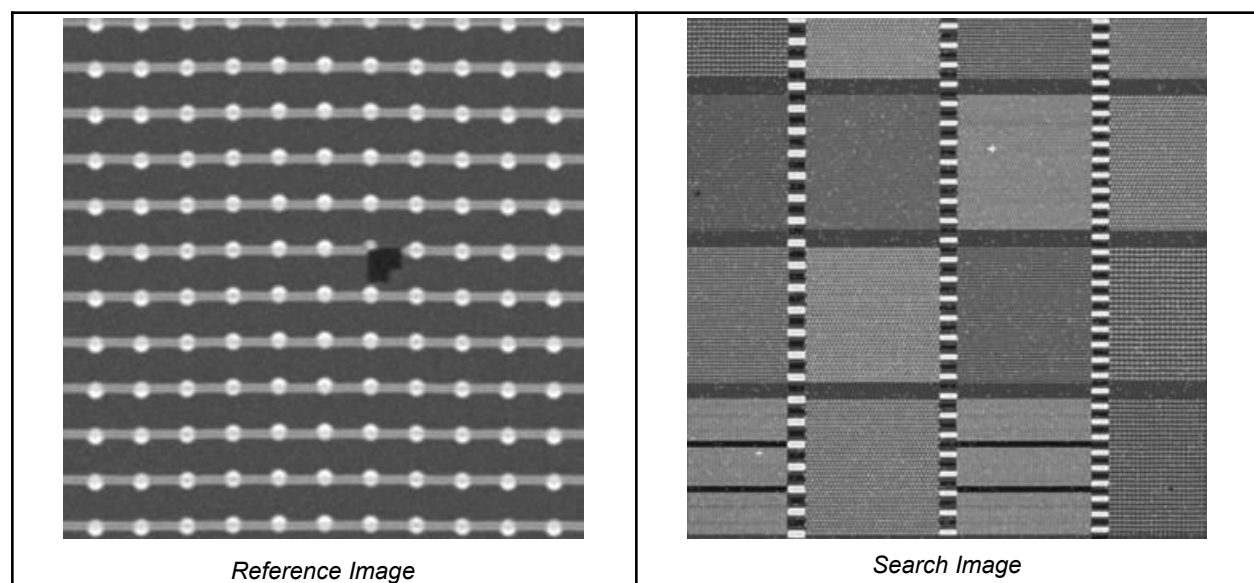
Difficulty Tier	Easy
Noise Variation (types)	salt_pepper + raster_drift + shot_noise + edge_brightening
Dose Scale	1.193
Shear	1.322 px
Jitter	0.813 px
Beam Spot	5.570 nm
Edge Strength	0.394
Architecture Type	DRAM (dram_staggered_realistic, wavy_dram_bitline) with a finfet_via_realistic companion style present in the same world

Noise Citations	salt pepper: Boyat, A.K. & Joshi, B.K. (2015). "A Review Paper: Noise Models in Digital Image Processing." <i>arXiv:1505.03489</i>. (Salt-and-pepper / impulse noise model) https://arxiv.org/pdf/1505.03489 raster drift: Sang, X. & LeBeau, J.M. (2016). "Correcting nonlinear drift distortion of scanning probe microscopy from image pairs with orthogonal scan directions." <i>Ultramicroscopy</i> 187, 98-108 (<i>arXiv:1507.00320</i>). https://arxiv.org/pdf/1507.00320 shot noise: Sim, K.S. et al. (2025). "Signal-to-Noise Ratio in Scanning Electron Microscopy: A Comprehensive Review." <i>arXiv:2510.07886</i>. https://arxiv.org/abs/2510.07886 edge brightening: Electron Microscopy ETH Zurich. "Secondary Electron Imaging — Edge Effect." <i>ETH Zurich Scientific Center for Optical and Electron Microscopy</i>. https://www.microscopy.ethz.ch/se.htm
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Pattern Uniqueness — Justification

Horizontal bitlines with periodic bright storage-node dots are interrupted by a single asymmetric dark L-shaped defect blob in one row — a one-off local landmark not repeated elsewhere in the dataset. This is combined with salt-and-pepper speckling, a noise component absent from samples 1 and 2.

Reference & Search Images



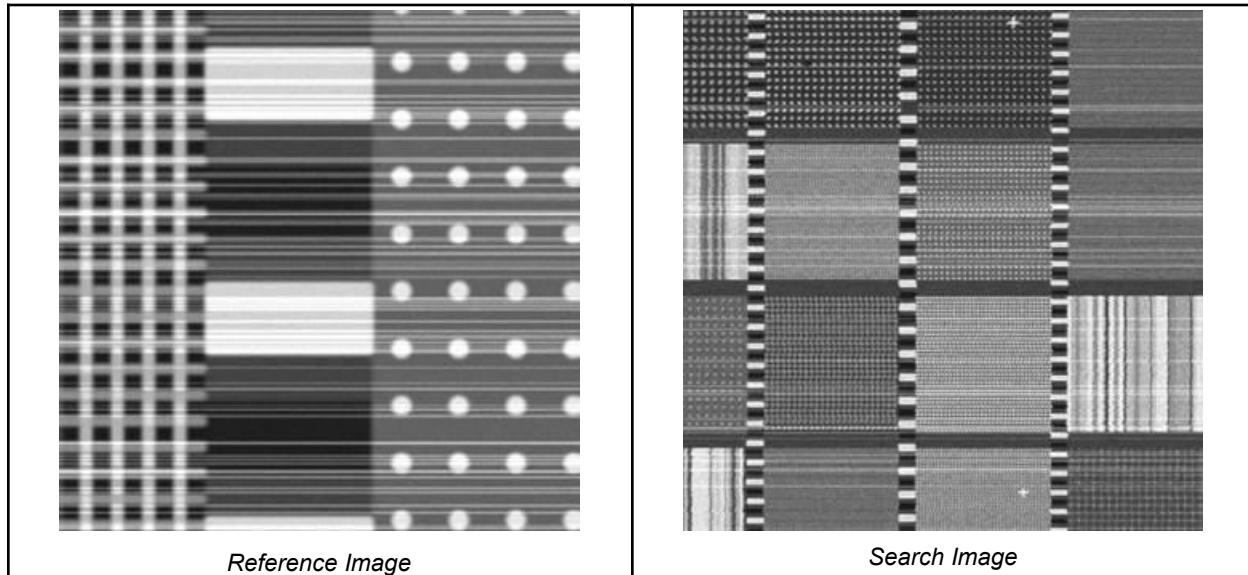
Sample 4 — sample_0856: Logic / BEOL Contact Array (Medium Noise)

Difficulty Tier	Medium
Noise Variation (types)	charging + speckle + raster_drift + shot_noise + edge_brightening
Dose Scale	0.778 (under-dosed / darker)
Shear	2.749 px
Jitter	1.334 px
Beam Spot	6.531 nm
Edge Strength	0.328
Architecture Type	Logic / BEOL (mesh_grid, contact_array_round, standard_cell_regular) — again non-DRAM/FinFET
Noise Citations	charging: Kim, J. et al. (2025). "Deep learning denoising enables rapid SEM imaging under charging conditions for FE SEM, CD SEM, and review SEM." <i>Scientific Reports</i>. https://www.nature.com/articles/s41598-025-33273-3 speckle: Boyat, A.K. & Joshi, B.K. (2015). "A Review Paper: Noise Models in Digital Image Processing." <i>arXiv:1505.03489</i>. https://arxiv.org/pdf/1505.03489 raster drift: Sang, X. & LeBeau, J.M. (2016). "Correcting nonlinear drift distortion of scanning probe microscopy from image pairs with orthogonal scan directions." <i>Ultramicroscopy</i> 187, 98-108 (<i>arXiv:1507.00320</i>). https://arxiv.org/pdf/1507.00320 shot noise: Sim, K.S. et al. (2025). "Signal-to-Noise Ratio in Scanning Electron Microscopy: A Comprehensive Review." <i>arXiv:2510.07886</i>. https://arxiv.org/abs/2510.07886 edge brightening: Electron Microscopy ETH Zurich. "Secondary Electron Imaging — Edge Effect." <i>ETH Zurich Scientific Center for Optical and Electron Microscopy</i>. https://www.microscopy.ethz.ch/se.htm

Pattern Uniqueness — Justification

This is the first sample in the set where 'charging' noise appears, visible as fine horizontal scan-line banding overlaid across all three sub-regions (crosshatch mesh, gray step-wedge, and round contact dots). This charging signature is not present in samples 1 through 3.

Reference & Search Images



Sample 5 — sample_1057: DRAM + FinFET (Legacy, Hard Tier, Full Noise Stack)

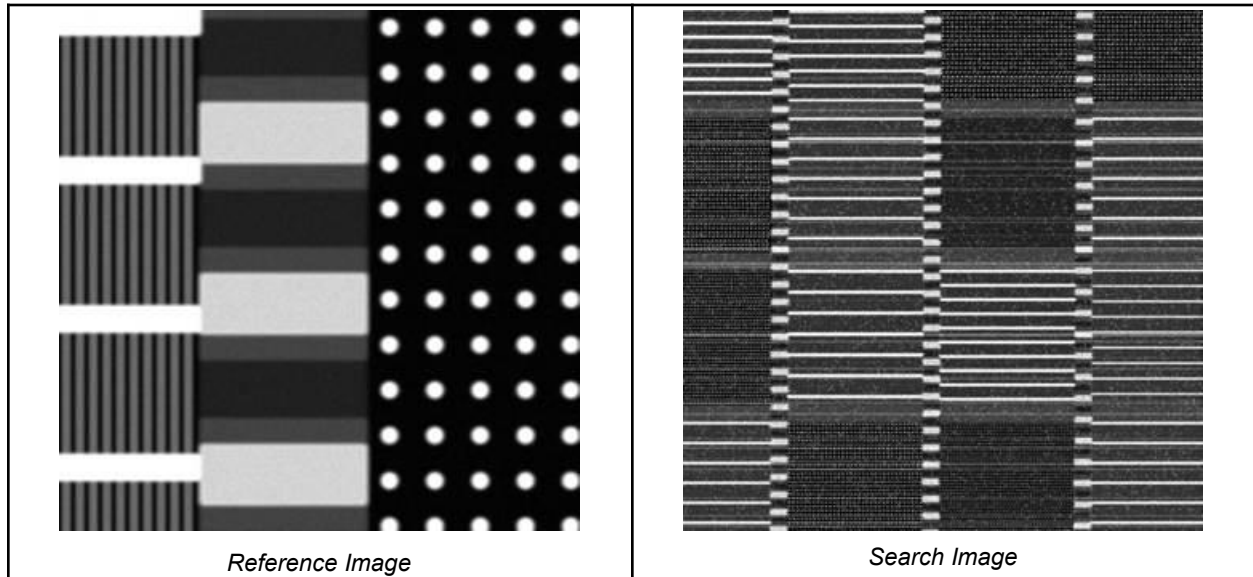
Difficulty Tier	Hard
Noise Variation (types)	charging + speckle + salt_pepper + raster_drift + shot_noise + edge_brightening (all 6 types)
Dose Scale	0.550 (heavily under-dosed)
Shear	3.844 px
Jitter	1.364 px
Beam Spot	7.762 nm
Edge Strength	0.295
Architecture Type	DRAM (dense dot capacitor array) + FinFET (vertical fins) — legacy_dram_1x + legacy_finfet

Noise Citations	charging: Kim, J. et al. (2025). "Deep learning denoising enables rapid SEM imaging under charging conditions for FE SEM, CD SEM, and review SEM." <i>Scientific Reports</i>. https://www.nature.com/articles/s41598-025-33273-3 speckle: Boyat, A.K. & Joshi, B.K. (2015). "A Review Paper: Noise Models in Digital Image Processing." <i>arXiv:1505.03489</i>. https://arxiv.org/pdf/1505.03489 salt pepper: Boyat, A.K. & Joshi, B.K. (2015). "A Review Paper: Noise Models in Digital Image Processing." <i>arXiv:1505.03489</i>. (Salt-and-pepper / impulse noise model) https://arxiv.org/pdf/1505.03489 raster drift: Sang, X. & LeBeau, J.M. (2016). "Correcting nonlinear drift distortion of scanning probe microscopy from image pairs with orthogonal scan directions." <i>Ultramicroscopy</i> 187, 98-108 (<i>arXiv:1507.00320</i>). https://arxiv.org/pdf/1507.00320 shot noise: Sim, K.S. et al. (2025). "Signal-to-Noise Ratio in Scanning Electron Microscopy: A Comprehensive Review." <i>arXiv:2510.07886</i>. https://arxiv.org/abs/2510.07886 edge brightening: Electron Microscopy ETH Zurich. "Secondary Electron Imaging — Edge Effect." <i>ETH Zurich Scientific Center for Optical and Electron Microscopy</i>. https://www.microscopy.ethz.ch/se.htm
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Pattern Uniqueness — Justification

Same material families as sample_0083, but pushed to maximum noise difficulty. The fin stripes show bright horizontal 'break' bands (wordline straps) not seen in sample_0083, and the DRAM dot pitch is denser and higher-contrast. This is the clearest illustration of how identical architecture classes look completely different once every noise channel (charging, salt-pepper, drift) stacks up simultaneously.

Reference & Search Images



Sample 6 — sample_1393: Logic / BEOL Via/Ring Array (Heaviest Degradation)

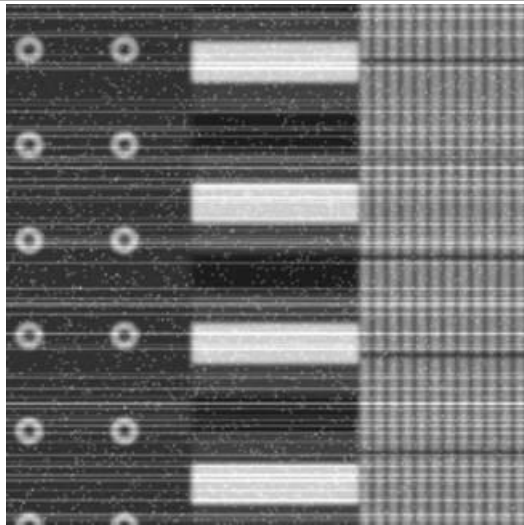
Difficulty Tier	Hard
Noise Variation (types)	Search: charging + speckle + raster_drift + shot_noise + edge_brightening. Reference: charging + salt_pepper + shot_noise + edge_brightening
Dose Scale	0.381 (lowest of all six)
Shear	5.180 px (largest drift)
Jitter	1.212 px
Beam Spot	8.149 nm (most blurred)
Edge Strength	0.201
Architecture Type	Logic / BEOL (ring_array / via rings, layered_cell, mesh_grid) — non-DRAM/FinFET

Noise Citations	charging: Kim, J. et al. (2025). "Deep learning denoising enables rapid SEM imaging under charging conditions for FE SEM, CD SEM, and review SEM." <i>Scientific Reports</i>. https://www.nature.com/articles/s41598-025-33273-3 speckle: Boyat, A.K. & Joshi, B.K. (2015). "A Review Paper: Noise Models in Digital Image Processing." <i>arXiv:1505.03489</i>. https://arxiv.org/pdf/1505.03489 salt pepper: Boyat, A.K. & Joshi, B.K. (2015). "A Review Paper: Noise Models in Digital Image Processing." <i>arXiv:1505.03489</i>. (Salt-and-pepper / impulse noise model) https://arxiv.org/pdf/1505.03489 raster drift: Sang, X. & LeBeau, J.M. (2016). "Correcting nonlinear drift distortion of scanning probe microscopy from image pairs with orthogonal scan directions." <i>Ultramicroscopy</i> 187, 98-108 (<i>arXiv:1507.00320</i>). https://arxiv.org/pdf/1507.00320 shot noise: Sim, K.S. et al. (2025). "Signal-to-Noise Ratio in Scanning Electron Microscopy: A Comprehensive Review." <i>arXiv:2510.07886</i>. https://arxiv.org/abs/2510.07886 edge brightening: Electron Microscopy ETH Zurich. "Secondary Electron Imaging — Edge Effect." <i>ETH Zurich Scientific Center for Optical and Electron Microscopy</i>. https://www.microscopy.ethz.ch/se.htm
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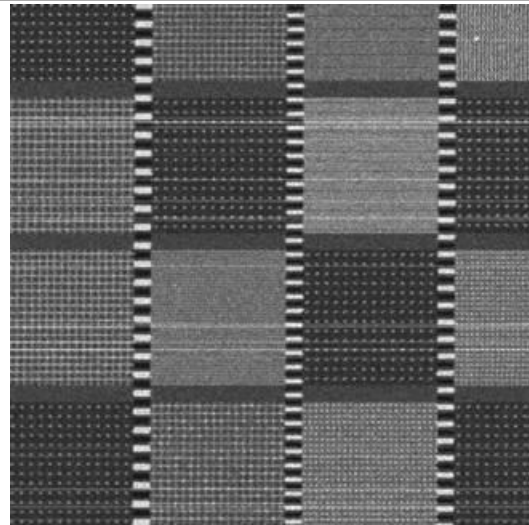
Pattern Uniqueness — Justification

Visibly the worst-degraded image in the set — dense white salt-pepper specks plus strong horizontal raster-drift lines wash across a ring-via array, a gray step-band region, and a fine vertical mesh simultaneously. It is the only sample where the REFERENCE image (not just the search image) carries salt-pepper noise, making it a distinct edge case for evaluating noise robustness.

Reference & Search Images



Reference Image



Search Image